A Mini Project 1 Report on

Mindscape-Mental Health Tracking Application

Submitted to the Department of Computer Science & Engineering, GNITS in the partial fulfillment of the academic requirement for the award of B.Tech (CSE)

Under JNTUH, Hyderabad

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Shaikpet, Hyderabad-5001004

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Certificate

This is to certify that the Mini-Project1 report on **"Mindscape-Mental Health Tracking Application."** is a bonafide work carried out by **B.Likhitha (22251A0567), D.Prasanna (22251A05A5), T.Meghana (23255A0510), M.Anusha (23255A0513)** in the partial fulfillment for the award of B.Tech degree in Computer Science & Engineering , G. Narayanamma Institute of Technology & Science, Shaikpet, Hyderabad, affiliated to Jawaharlal Nehru Technological University, Hyderabad under our guidance and supervision for the academic year 2023-2024.

The results embodied in the Mini-Project1 work have not been submitted to any other University or Institute for the award of any degree or diploma.

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Abstract

Mental health plays a vital role in our overall well-being. It is just as important as our physical health and, as a result, heavily influences our quality of life. In today's context, where stressful work environments, social isolation, and academic pressure are prevalent, mental health has emerged as a significant concern. Those who are struggling with their mental health are more likely to experience relationship challenges, physical health issues, and disruptions in their long-term goals. Statistics show that 1 in every 5 individuals suffers from some form of mental health illness symptoms. Unfortunately, most people who struggle with their mental health do not receive the help they deserve. It is a growing concern as the most affected section of the population with mental health issues are students and youngsters.

The proposed model aims to create a user-friendly mental health tracker. With the use of this application, people will be able to discreetly evaluate their mental health and get personalized suggestions for betterment. Developing a user interface that is both aesthetically pleasing and easy to use, putting in place a minimally intrusive evaluation procedure, producing tailored suggestions based on user input, and safely preserving records of users mental health status and advancement are among the key goals. In order to guarantee a seamless user experience across a range of devices and network circumstances, the app will also give priority to accessibility and speed optimization. By offering a private setting for self-evaluation and tailored advice, users can handle developing mental health issues before they get worse. Additionally, the project contributes to reducing the stigma associated with mental illness and promotes awareness and destigmatization of mental health issues. Ultimately, by prioritizing mental health and well-being, the project seeks to empower individuals to lead healthier, more fulfilling lives.

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1. INTRODUCTION

1.1. Background of the study

Mental health is a fundamental component of overall well-being, influencing our thoughts, emotions, and behaviors. It affects how we handle stress, relate to others, and make choices. Mental health issues are prevalent in today's society, with stress, anxiety, and depression being among the most common problems. When mental health is compromised, it can affect all aspects of life, including physical health, social connections, and productivity. Therefore, maintaining mental health and addressing mental health issues promptly is vital for a healthy, balanced life. Given the rising prevalence of mental health issues, there is a growing need for effective tools to manage mental health. Traditional methods of mental health care, such as in-person therapy and counseling, can be effective but often face barriers like accessibility, cost, and stigma. Various tools and applications aim to address these mental health issues, each with its own strengths and limitations. While wearable devices can provide valuable insights, they often come with higher costs and require users to wear them consistently, which can be inconvenient. Mood trackers help users log their moods over time and journaling apps allow users to record their thoughts and feelings, providing an outlet for self-expression and reflection. However, these existing systems often have limitations such as time consumption, lack of comprehensive support, and the inability to consistently monitor mental health.

There is a pressing need for an effective mental health tracking application that is simple to use, integrates seamlessly into daily life, and enables consistent monitoring of mental health arising from several critical factors in contemporary society. Integrating such applications into daily life ensures that mental health monitoring becomes a natural part of one's routine, promoting consistent usage and long-term adherence. Ultimately, an accessible and effective mental health tracking application helps empower individuals to take control of their mental well-being.

1.2. Problem Statement

A reliable monitoring tool is crucial to effectively track and manage these concerns. Existing solutions often fail to provide comprehensive support tailored to individual needs, resulting in limited engagement and suboptimal outcomes. There is a pressing need for an innovative app that empowers well-being, receives personalized support, and gains valuable insights through data-driven analysis, ultimately fostering improved mental health outcomes and well-being.

1.3. Existing Systems

Several mental health applications are available to support mental well-being. Moodpath utilizes machine learning to help users understand their moods and triggers, offering personalized recommendations and coping mechanisms. Daylio enables users to track their mood, activities, and thoughts throughout the day, with features to set goals and monitor progress over time. Day One is a user-friendly journaling app that allows users to enrich entries with photos, videos, and audio recordings. Journey provides various journaling prompts and exercises, facilitating ease in starting, and includes tools to track mood and set personal goals.

1.4. Advantages & Drawbacks

Advantages:

- Combining mood tracking with daily activity and thought logging (as seen in Daylio) provides a comprehensive understanding of how daily life affects mental health. This holistic approach identifies multiple factors contributing to emotional well-being, enabling targeted and effective coping strategies.
- Moodpath uses machine learning to offer highly personalized insights and recommendations by analyzing vast amounts of user data.

- Apps like Dayone and Journey enhance journaling with multimedia elements and varied prompts, making it more engaging. This approach allows for diverse self-expression and reflection.

Drawbacks:

- Mood tracker app can be time-consuming and may not be accurate if you don't use them consistently.
- Journaling apps can be time-consuming and may not be for everyone. Some people may find it difficult to be honest with themselves when journaling.
- Meditation apps may not be effective for everyone. Some people may find it difficult to clear their minds or focus on their breath.

1.5. Proposed System

The proposed model adopts a two-question-a-day approach to mental health tracking, making the process straightforward and manageable for users. By integrating established psychological scales such as the Perceived Stress Scale (PSS) and the Generalized Anxiety Disorder (GAD) scale, it offers a comprehensive evaluation of mental well-being. Based on user responses, the model provides personalized recommendations, delivering tailored tips for stress management and relaxation, thereby enhancing the overall user experience and promoting better mental health.

1.6. Methodology

The “Mindscape-mental health tracking application” involves several critical steps to ensure effective monitoring and support for mental health. The app prompts users with daily questions to track their mental well-being. Users respond to questions designed to assess their stress and anxiety levels using established psychological scales like the Perceived Stress Scale (PSS) and Generalised Anxiety Disorder (GAD-7).

Upon logging in, users answer two simple questions daily. The app uses these responses to calculate stress and anxiety scores based on the PSS and GAD-7 criteria. Each response is stored securely in a Firebase real-time database, ensuring data privacy and security. The app then processes this data using cloud functions to generate personalized recommendations. These recommendations aim to help users manage their mental health by providing tips and strategies tailored to their specific needs.

If a user’s scores indicate consistently high levels of stress or anxiety, the app triggers an intervention by suggesting the user to seek professional help. This ensures that users who may need additional support receive timely and relevant recommendations. The user interface, developed with Flutter, is designed to be intuitive and engaging, promoting regular use and long-term engagement.

The backend infrastructure, powered by Firebase, handles user authentication, data storage, and cloud functions seamlessly, ensuring a smooth user experience. By integrating these technologies, the app offers a simple yet effective solution for mental health tracking, fostering consistent and long-term user engagement while maintaining a high standard of data security.

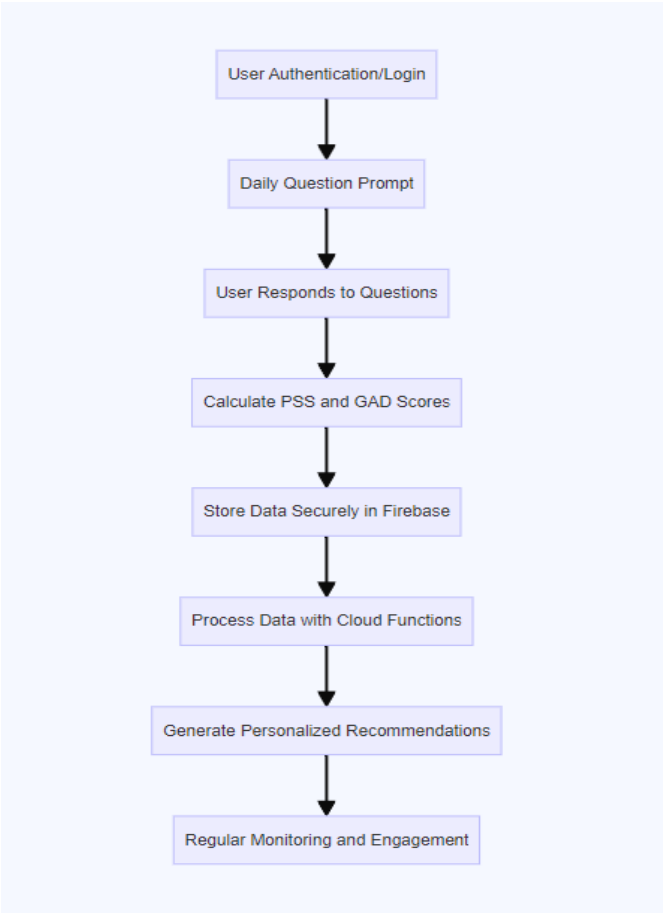

Fig 1.6. Methodology for Mindscape

1.7. Objectives of the project

1. To develop a user-friendly mental health tracking application that is hassle-free.
2. To integrate established psychological scales and personalized recommendations to enhance user engagement.
3. To foster long-term mental well-being habits by promoting regular self-assessment and stress management skills among users.

1.8. Organization of the Project

The proposed model adopts a two-question-a-day approach to mental health tracking, making the process straightforward and manageable for users. This work is organized into six chapters. Chapter1 gives a complete introduction of the system including the background study, existing systems, the proposed system of the project, objectives, and methodology.

Chapter 2 is regarding the literature survey conducted for the project.

Chapter 3 elaborates on the architecture and module design of the Mindscape.

Chapter 4 describes the implementation of the system. The implementation includes detailed description of the implementation of each module, and the working.

Chapter 5 shows the results obtained and screenshots of the application user interface showing results.

Chapter 6 describes work done and future scope.

Chapter 7 shows the references used to build the project.

2. LITERATURE SURVEY:

Mendis E.S. and Samarasimha L., 2022[1], did research on Mobile Applications for mental Health using Machine Learning. This study focuses on a mobile app utilizing machine learning algorithms like LSTM and BERT to monitor mental health. The app achieves an accuracy of 64-70%. Key limitations include data privacy concerns and the need for consistent user engagement. The effectiveness of the app largely depends on the quality and consistency of user data. Despite these challenges, the study demonstrates the potential of machine learning in enhancing mental health support through mobile technology.

Malaika Samuel, C.P. Shirley , 2023[2], did research on Mindset: Mental Well-being Support application.This presents an Android-based app designed to support mental well-being using cognitive-behavioral therapy (CBT) principles. The app offers tools and resources like CBT exercises and personalized recommendations, achieving an accuracy of 70-85% when used consistently. A noted limitation is that the app can be time-consuming and less effective without regular use. Despite this, "Mindset" aims to provide a comprehensive and accessible solution for mental well-being through its user-friendly interface and evidence-based approach.

C.Jyothsna, S. Amudha, Amritanshu Ram, 2023[3], did research on PredictEye.This uses machine learning to predict mental states via webcam data. It employs Random Forest and LSTM algorithms, achieving an accuracy rate of 86.4%. The system provides real-time mental state predictions but faces data privacy and security concerns due to the sensitive visual data it collects. Despite these challenges, PredictEye offers a promising tool for early detection and intervention in mental health through the combination of machine learning and accessible webcam technology.

Year	Project	Authors	Technology	Algorithms /Tools	Accuracy	Limitations
2022	Mobile Applications for mental health using machine learning	Mendis E.S. Samarasimha L.	Machine Learning	LSTM (or)	Varied (64-70%)	Data Privacy, user Engagement concerns
2022	StressNet: Deep Learning For Stress Detection	Lee et al	Smartphone applications, Physiological sensors	CNN, LSTM	Varied (70-85%)	Environmental factors, Data privacy and security
2023	Mindset: Mental Well-being Support application	Malaika Samuel, C.P. Shirley	Android FrameWork	Cognitive Behavioural Therapy (CBT)	Varied (70-85%)	Time-Consuming, Less accurate Without Consistent use.
2023	Predict Eye	C. Jyothsna S. Amudha, Amritanshu Ram	Machine Learning	Random Forest Algorithm, Long-Short Term Memory (LSTMA)	Varied (86.4%)	Data privacy and security
2023	Predictive Models for Early Detection Of Depression From Smartphone Data	A. Smith, Et al	Smartphone Sensors and usage data	Ensemble learning methods, Feature engineering	Varied (80%)	Data privacy Concerns, Variability in user behaviour Affecting Model robustness.

3. MINDSCAPE-MENTAL HEALTH TRACKING APPLICATION

The proposed model utilizes a simple and effective method for mental health tracking by asking users two questions each day. This approach makes the process easy to manage. By incorporating well-known psychological scales like the Perceived Stress Scale (PSS) and the Generalized Anxiety Disorder (GAD) scale, the model ensures a thorough assessment of mental well-being. User responses are then used to offer personalized recommendations, including stress management and relaxation tips, which enhance the user experience and support improved mental health.

3.1. Architecture of the System:

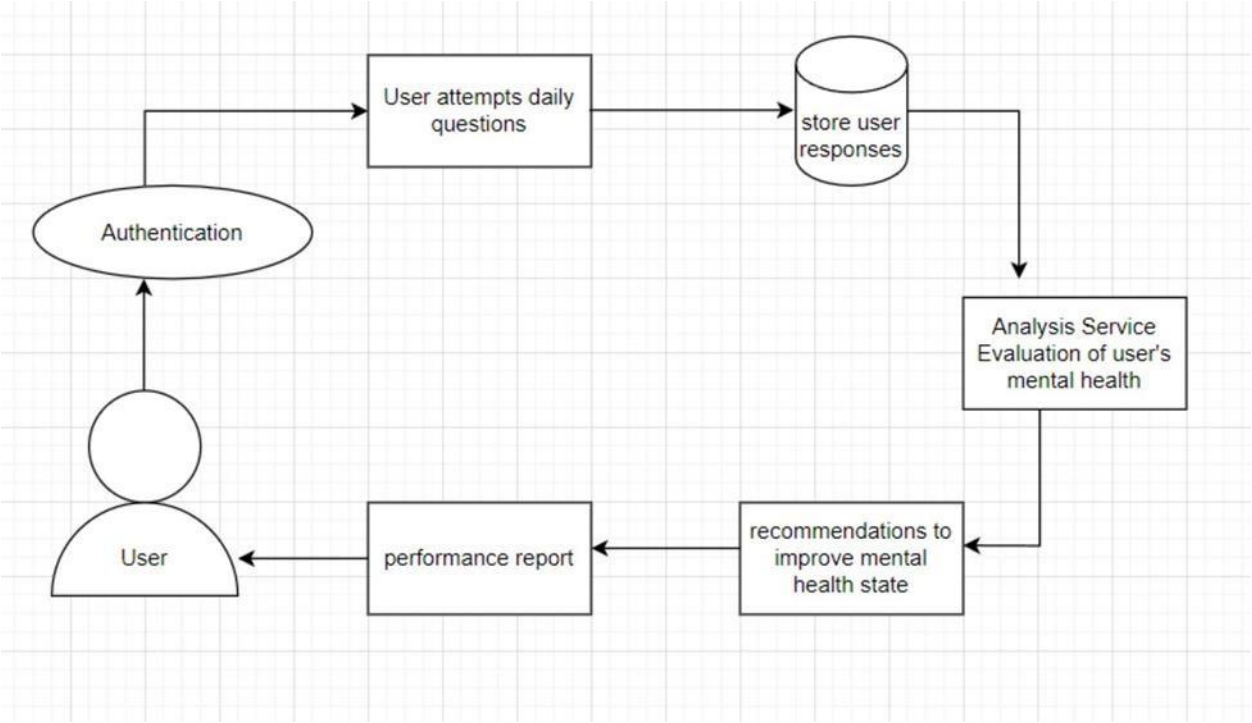

Fig 3.1. Describes the Architecture of the Project

The architecture for the mental health tracking application is designed to provide a user-friendly and comprehensive solution for mental well-being. The frontend of the application is built using the Flutter SDK, a cross-platform development framework that enables the creation

of high-performance apps for web, iOS and Android. This framework facilitates the development of user input forms for daily questions, interfaces for viewing personalized recommendations, and tracking progress over time.

On the backend, the application uses the Firebase SDK, which offers a variety of services essential for mobile and web app development, including a real-time database, authentication, and cloud functions. The real-time database stores user responses and progress data, while the authentication service ensures secure user login and data protection. Cloud functions handle the processing of user data and the generation of personalized recommendations.

Data processing in the application leverages established psychological scales such as the Perceived Stress Scale (PSS) and the Generalized Anxiety Disorder (GAD) scale, along with machine learning models. These tools analyze user responses and predict mental health trends, enabling the provision of tailored tips for stress management and relaxation based on individual data.

The hardware requirements for the application include a multicore processor (Intel Core i5 or above) and 8 GB of RAM, ensuring smooth operation and performance. The primary objectives of the system are to develop a hassle-free and user-friendly mental health tracking application, integrate comprehensive psychological evaluations, and enhance user engagement through personalized recommendations. By promoting regular self-assessment and long-term well-being habits, the architecture aims to provide a robust, secure, and effective system for monitoring and improving mental health.

3.2. Module Design

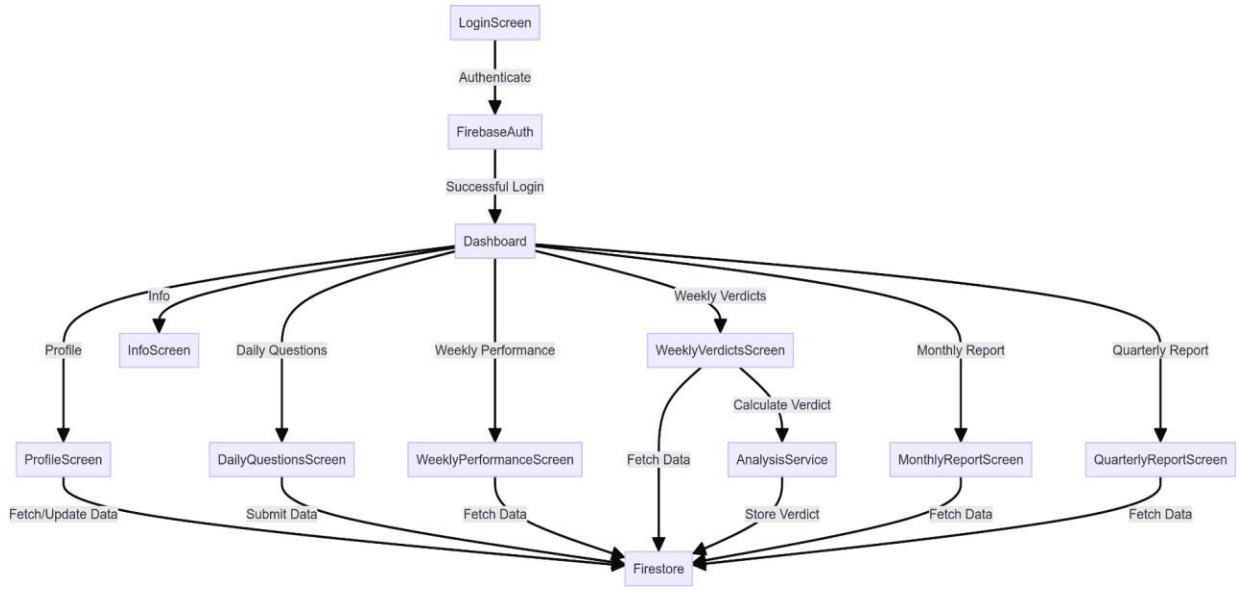

Fig 3.2 Module Designing

3.2.1. User Interface

The User Interface (UI) module is developed using the Flutter SDK to provide a seamless experience across both iOS and Android platforms. It includes interfaces for daily questions, progress tracking, and personalized recommendations. The design focuses on simplicity and ease of use to encourage consistent user engagement.

3.2.2. Backend Module

The Backend module employs Firebase for robust backend services. It handles user authentication, data storage, and cloud functions. This module ensures secure login processes, real-time data storage and retrieval, and efficient execution of server-side functions to process user data and generate recommendations.

3.2.3. Data Processing

The Data Processing module is responsible for analyzing user responses. It integrates established psychological scales such as PSS and GAD to evaluate mental well-being. Additionally, machine learning models analyze patterns in the data to predict mental health trends, providing a deeper understanding of user needs.

3.2.4. Recommendations

The Recommendations module uses the insights gained from the data processing module to provide personalized tips for stress management and relaxation. It dynamically updates recommendations based on ongoing user input and progress, ensuring that the advice is relevant and effective.

4. IMPLEMENTATION OF MINDSCAPE USING FLUTTER AND FIREBASE

4.1. User Authentication/Login:

The user begins by logging into the app, utilising Firebase Authentication for secure and reliable access. This ensures that user data is protected and accessible only to authenticated individuals. The credentials used for login are an email and a password, providing a straightforward and familiar method for users to access their accounts.This approach not only simplifies the login process but also leverages Firebase's robust security features to safeguard user information**.**

4.2. User Response Collection:

Upon logging in, the user is presented with a daily question prompt. The questions are designed to gauge the user's current stress and anxiety levels. Users respond to the daily questions, and their answers are immediately captured by the app. This seamless data collection process ensures that users can quickly provide their inputs and continue with their day.

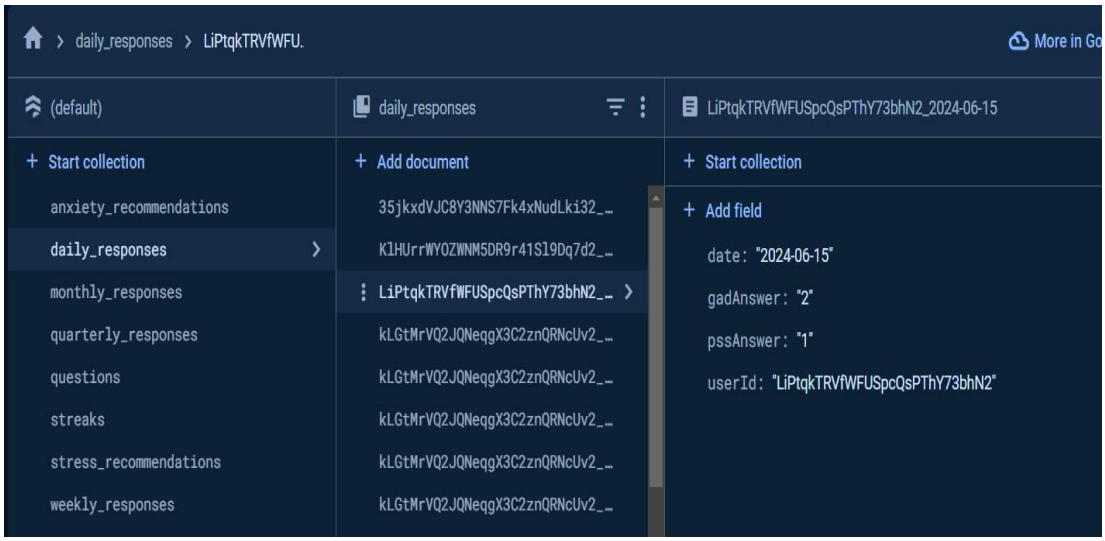

Fig 4.2 User Response Dataset

4.3. Calculation of PSS and GAD Scores:

4.3.1. Perceived Stress Scale

The Perceived Stress Scale (PSS) is a psychological instrument designed to measure the perception of stress. It assesses how unpredictable, uncontrollable, and overloaded respondents find their lives. The scale is widely used because it captures the subjective experience of stress, rather than just objective stressors.

Scoring Criteria:

- The PSS consists of 10 items (PSS-10), though there are versions with 4 (PSS-4) and 14 (PSS-14) items.
- Respondents indicate how often they felt or thought a certain way during the last month using a 5-point scale ranging from "0 = Never" to "4 = Very Often."
- Positively stated items (e.g., items 4, 5, 7, and 8 on the PSS-10) are reverse scored. For example, if a respondent selects "4 = Very Often" for a positively stated item, it is scored as "0."
- The scores are summed to create an overall stress score. Higher scores indicate higher perceived stress.

Interpreting Scores:

- Low stress: 0-13
- Moderate stress: 14-26
- High stress: 27-40

4.3.2. Generalised Anxiety Disorder (GAD-7)

The Generalised Anxiety Disorder (GAD-7) scale is a brief self-report questionnaire used to screen for and measure the severity of generalised anxiety disorder. It is commonly used in clinical and research settings.

Scoring Criteria:

- The GAD-7 consists of 7 items.
- Respondents rate how often they have been bothered by each symptom over the last two weeks using a 4-point scale ranging from "0 = Not at all" to "3 = Nearly every day."
- The scores for each of the 7 items are summed to create a total score.

Interpreting Scores:

- Minimal anxiety: 0-4
- Mild anxiety: 5-9
- Moderate anxiety: 10-14
- Severe anxiety: 15-21

```
  // Calculate the average GAD and PSS scores
  double totalGadScore = 0;
  double totalPssScore = 0;
  int count = dailyResponsesSnapshot.docs.length;

  for (var doc in dailyResponsesSnapshot.docs) {
    totalGadScore += double.parse(doc['gadAnswer']);
    totalPssScore += double.parse(doc['pssAnswer']);
  }

  double averageGadScore = totalGadScore / count;
  double averagePssScore = totalPssScore / count;

  // Determine the verdict based on the scores
  String verdict = _determineVerdict(averageGadScore, averagePssScore);

  // Store the weekly verdict
  await _firestore.collection('weekly_verdicts').add({
    'userId': userId,
    'startOfWeek': DateFormat('yyyy-MM-dd').format(startOfWeek),
    'endOfWeek': DateFormat('yyyy-MM-dd').format(endOfWeek),
    'averageGadScore': averageGadScore.toString(),
    'averagePssScore': averagePssScore.toString(),
    'verdict': verdict,
  });

  print('Weekly verdict has been created and stored successfully');
} catch (e) {
  print('Error creating weekly verdict: $e');
}
```

Fig 4.3 Calculating the Scores

4.4. Generation of Personalised Recommendations:

Based on the calculated PSS and GAD scores, the app generates personalised recommendations for the user with the help of fetch Recommendations method.These recommendations include practical tips and techniques for managing stress and anxiety, tailored to the individual's current mental health status.

```
Future<String> fetchRandomRecommendation(String collectionName) async {
  try {
    QuerySnapshot querySnapshot = await _db.collection(collectionName).get();
    List<QueryDocumentSnapshot> docs = querySnapshot.docs;
    if (docs.isEmpty) {
      throw Exception("No recommendations available");
    }

    var rng = Random();
    DocumentSnapshot randomDoc = docs[rng.nextInt(docs.length)];
    return randomDoc['recommendation'] ??
        'No recommendation available'; // Default message
  } catch (e) {
    print("Error fetching random recommendation: $e");
    rethrow;
  }
}
```

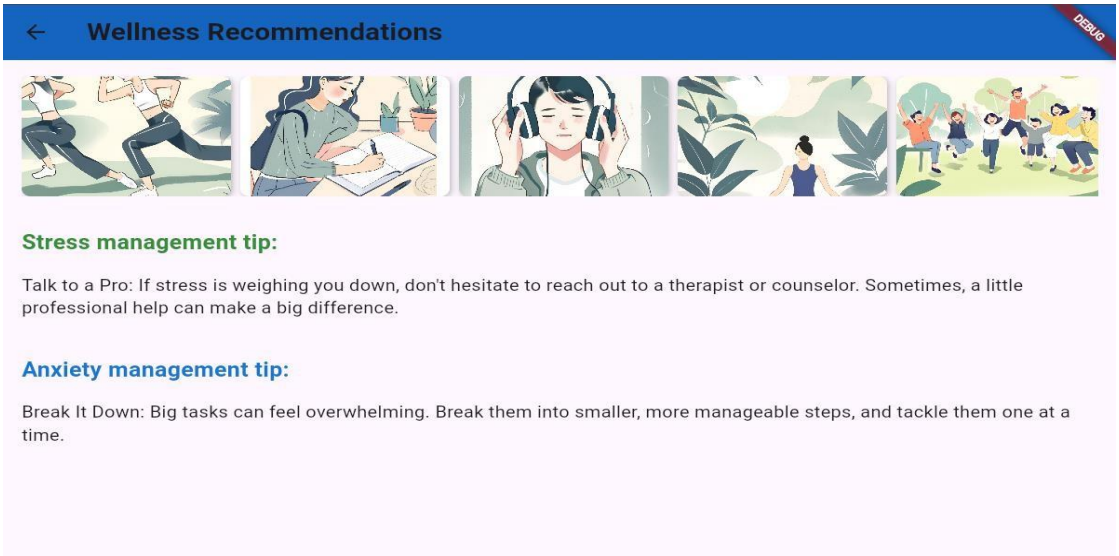

Fig 4.4 Wellness Recommendations

4.5. Technologies Used:

The system requirements for the project include both hardware and software specifications. The hardware must have a multicore processor (Intel Core i5 or above) and at least 8 GB of RAM for smooth performance. For software, the Flutter SDK is needed for frontend development. Flutter supports cross-platform development for iOS, Android, and web, featuring a single codebase, high-performance rendering, pre-designed widgets, and strong community support. For the backend, the Firebase SDK is required. Firebase provides real-time databases, authentication, cloud functions, and analytics, facilitating mobile and web app development. These requirements ensure robust and efficient application development.

5. RESULTS AND DISCUSSIONS

5.1. Results

The mental health tracking application project, referred to as Mindscape, has achieved significant milestones and demonstrated promising results:

- **User Engagement:** Mindscape has been designed to be user-friendly, integrating personalized daily and weekly interactions. This approach has successfully engaged users, promoting consistent use of the application.
- **Personalized Support:**By leveraging user data, the application provides tailored advice, helping users understand and manage their anxiety and stress levels effectively. This personalized approach has been a key factor in the app's success.
- **Secure and Seamless Experience:**The integration of Firebase for user authentication and data management ensures a secure and seamless user experience. Real-time data updates keep users informed and engaged with the latest information relevant to their mental health.
- **Intuitive User Interface:**The UI, developed using Flutter, promotes long-term engagement by being responsive and user-friendly. The dashboard design facilitates easy navigation and access to important features and data.
- **Effective Data Analysis:** Algorithms used in the app analyze user inputs to provide weekly mental health verdicts and personalized tips. This has helped used receive meaningful insights into their mental health status and actions to improve it.

5.2. Output Screens

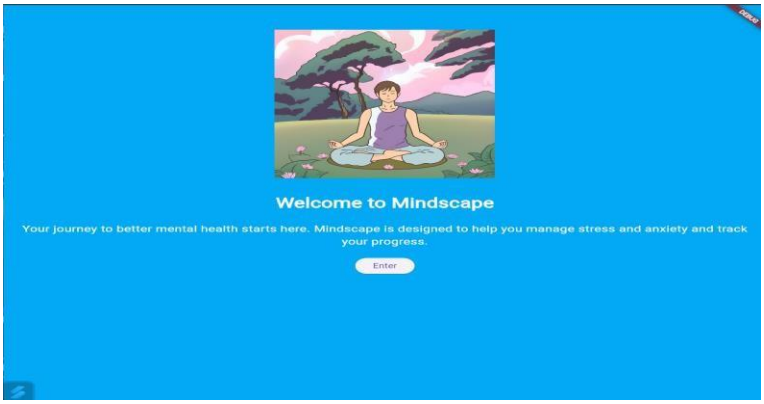

Fig 5.1 Home Screen

As shown in Figure 5.1, Mindscape welcomes interested users to embark on a journey toward better mental health. The platform aids in managing and eliminating stress & anxiety by tracking their mental well-being.

Fig 5.2 Login/Register

As shown in Figure 5.2, on the Mindscape login or register screen, new users can register, while returning users can log in to check their status and continue their recovery journey.

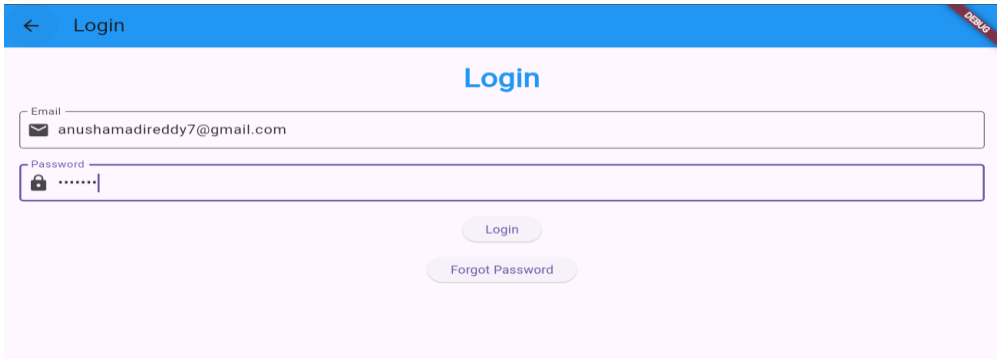

Fig 5.3 User Login

As shown in Figure 5.3, the Mindscape login page allows users to securely access their profiles, manage their mental health journey, update personal details, and track progress seamlessly.

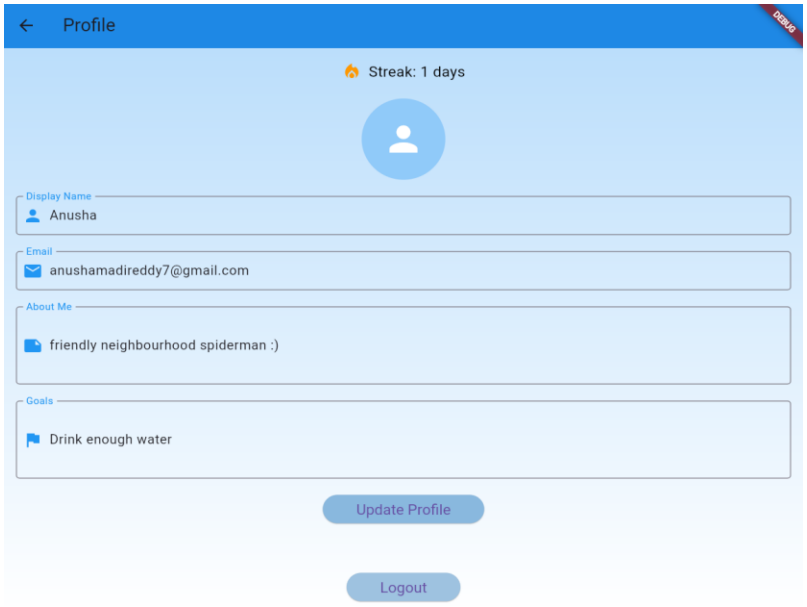

Fig 5.4 Profile Updation

As shown in Figure 5.4, On the Mindscape login page,users can access their profiles to update their username, password, the “About Me” section, and set personal goals for their recovery journey.

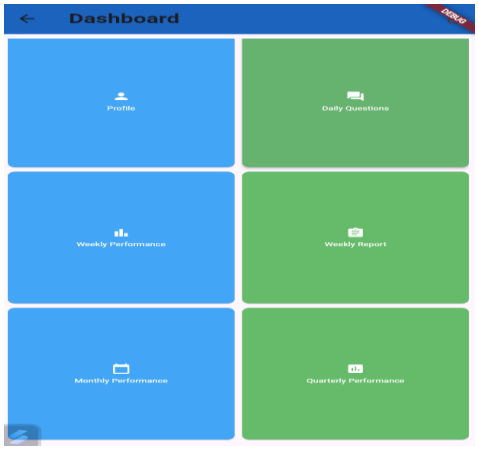

Fig 5.5 Dashboard

As shown in Figure 5.5, the Mindscape dashboard offers us various options, including profile updates, daily mood, anxiety and stress assessments, and performance tracking. Weekly reports, along with monthly and quarterly performance reviews, help users keep track of their mental health.

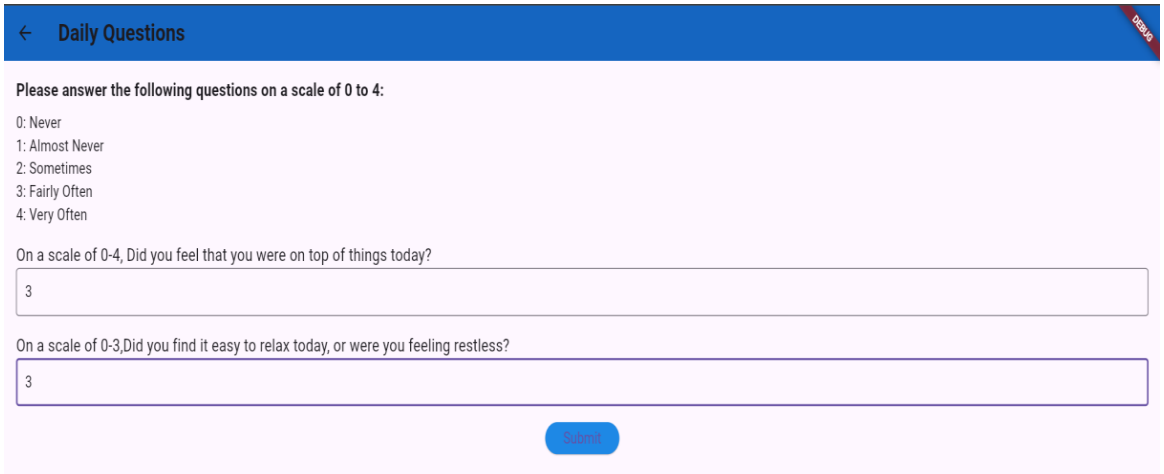

Fig 5.6 Daily Questions

As shown in Figure 5.5, after selecting the daily questions option, users will be prompted to answer stress and anxiety-assessing questions. They will rate their stress level on a scale of 0-4 and their anxiety level on a scale of 0-3.

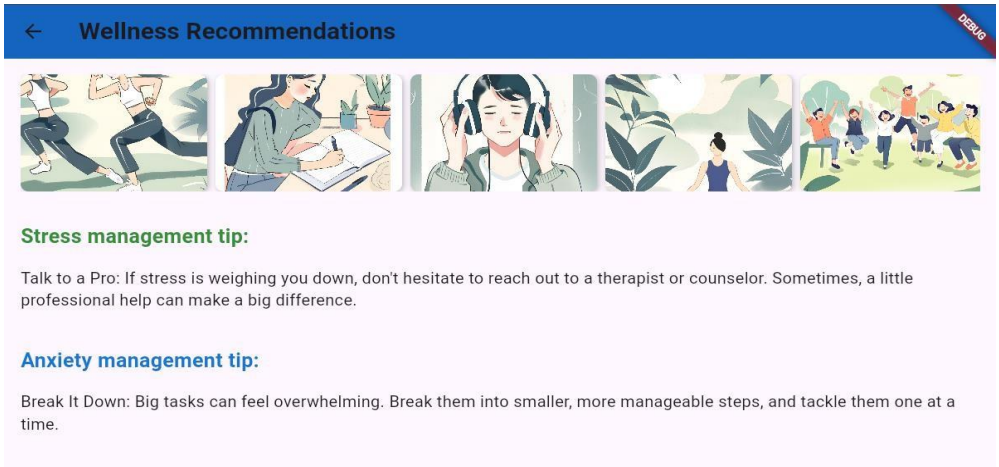

Fig 5.7 Wellness Recommendations

As shown in Figure 5.6, after completing the daily questions, users are immediately directed to the "Wellness Recommendation" screen. Here, they receive tailored stress and anxiety management tips based on their quiz responses.

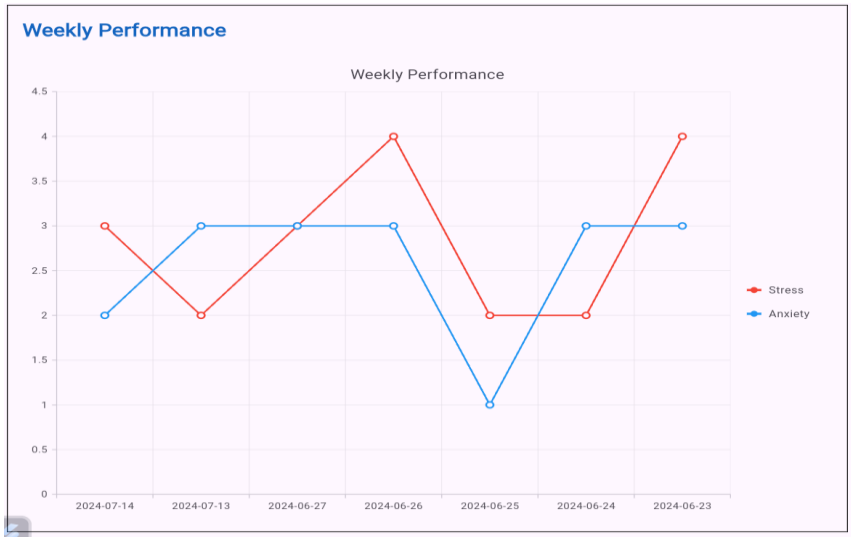

Fig 5.8 Weekly Performance graph

As shown in Figure 5.7, users can click on "Weekly Performance" to view their stress and anxiety levels analyzed in a graphical format. This helps users track their mental health progress over the week.

Fig 5.9 Weekly Report

As shown in Figure 5.8, users can access the "Weekly Report" after viewing the weekly performance graph. This report displays average GAD and PSS scores from all weekly tests, enabling users to assess their performance and track progress.

Fig 5.10 Generating monthly reports

As shown in Figure 5.9, additionally, users can access "Monthly Reports" to review their mental health progress over the month, providing a comprehensive view of trends and improvements.

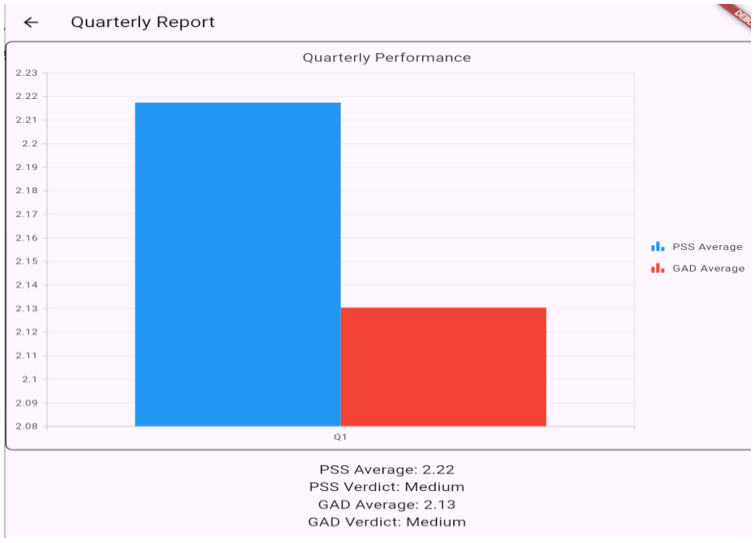

Fig 5.11 Generating Quarterly Report

As shown in Figure 5.10, the "Quarterly Reports" provide a detailed view, including average GAD and PSS scores and PSS and GAD verdicts, offering a comprehensive assessment of the user's mental health over three months.

6. CONCLUSIONS AND FUTURE ENHANCEMENTS

Conclusion

The mental health tracking application "Mindscape" represents a comprehensive and user-centric approach to monitoring and improving mental well-being. This project integrates modern technologies and psychological methodologies to offer a seamless, efficient, and engaging experience for its users.

Mindscape successfully addresses the need for accessible, effective, and personalized mental health tracking and support. By combining evidence-based psychological scales with modern technology, the application empowers users to take control of their mental health, providing them with the tools and insights necessary to make informed decisions and improve their well-being. The project's focus on user engagement, data security, and scalability ensures that Mindscape is well-positioned to adapt and grow, meeting the evolving needs of its users and making a meaningful impact in the field of mental health care.

Future Enhancements

- Implement machine learning algorithms to provide more accurate and personalized mental health insights.
- Analyze long-term user data to identify patterns and provide predictive mental health support.
- Develop a broader range of mental health tips and resources, including videos, articles, and interactive content.
- Partner with mental health professionals to ensure the accuracy and relevance of the content.
- Introduce community support features such as forums, group activities, and peer support systems.
- Facilitate connections with mental health professionals for online consultations.

REFERENCES

[1] Cohen, S., Kamarck, T., & Mermelstein, R. "A global measure of perceived stress," Journal of Health and Social Behavior, vol. 24, no. 4, pp. 385-396, 1983.

[2] Spitzer, R. L., Kroenke, K., Williams, J. B. W., & Löwe, B. "A brief measure for assessing generalized anxiety disorder: the GAD-7," Archives of Internal Medicine, vol. 166, no. 10, pp. 1092-1097, 2006.

[3] Mendis, E. S., Kasthuriarachchi, L. W., Samarasinha, H. P. K. L., Kasthuriarachchi, S., & Rajapaksa, S. "Mobile Application for Mental Health Using Machine Learning," 2022 4th International Conference on Advancements in Computing (ICAC), Colombo, Sri Lanka, pp. 387-392, 2022.

[4] Samuel, M., & Shirley, C. P. "Mindset, An Android-Based Mental Wellbeing Support Mobile Application," 2023 3rd International Conference on Pervasive Computing and Social Networking (ICPCSN). Salem, India, pp. 989-996.2023.doi:10.1109/ICPCSN5887.2023.00169.33211111.

[5] Jyotsna, C., Amudha, J., Ram, A., Fruet, D., & Nollo, G. "PredictEYE: Personalized Time Series Model for Mental State Prediction Using Eye Tracking," IEEE Access, vol. 11, pp. 128383-128409, 2023, doi: 10.1109/ACCESS.2023.3332762.

[6] Chitale, V., Playne, D., Liang, H.-N., & Baghaei, N. "Virtual Reality Data for Predicting Mental Health Conditions," 2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct),2022.

[7] Sharada, A., Ratkal, B., Lalitha, M., Samhitha, K.N.V., Rishika Reddy, A.(2023). Stress Detection in Classroom Environment Using Physiological Data. In: Seetha, M., Peddoju, S.K., Pendyala, V., Chakravarthy, V.V.S.S.S. (eds) Intelligent Computing and Communication. ICICC 2022. Advances in Intelligent Systems and Computing, vol 1447. Springer, Singapore. https://doi.org/10.1007/978-981-99-1588-0_15

GLOSSARY

- **Recommendation Engine:** A component of the application that users analysis results to generate personalized tips and insights for stress and anxiety management.
- **Resource Database:** Stores articles, videos, and interactive exercise related to Mental health, providing users with a broad range of resources and engaging content.
- **User Profile Dataset:** Contains basic information about the users, including their Unique identifiers, usernames, email addresses, and account creation dates.
- **Performance Metrics Dataset:** Tracks the computed metrics and summaries of users' Mental health status over time, including weekly stress and anxiety scores.
- **Real-Time Updates:** Refers to the dashboard updates that occur in real-time,ensuring Users have access to the most recent data and insights regarding their mental health.

APPENDIX

- **PSS (Perceived Stress Scale):** A psychological instrument used to measure the perception of stress. It consists of 10 questions designed to assess how unpredictable, uncontrollable, and overloaded respondents find their lives.
- **GAD-7 (Generalised Anxiety Disorder):** A seven-item scale used to measure the severity of generalised anxiety disorder